RareSense.AI

LLM-Powered Rare Disease Detection from Unstructured Clinical Notes

Review 1 — Project Overview, Data Sources & Case Study

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Date: June 2026

Domain: Health Data Science

Syllabus Ref: Project #7 (CareerPulse)

1. Problem Statement

Rare diseases affect over **300 million people worldwide**. Despite this, the average rare disease takes **7.6 years** to diagnose. Patients visit an average of **7 specialists** before receiving a correct diagnosis. The root cause: symptoms are scattered across years of clinical notes from different doctors, and no single physician sees the complete picture. Traditional keyword search fails because doctors use varied medical terminology for the same condition.

2. Our Solution: RareSense.AI

An AI system that **reads all of a patient's clinical notes**, extracts symptoms/diagnoses/medications using Clinical NLP, builds a unified patient timeline, and **cross-references the full symptom picture against 7,000+ rare diseases** to flag potential undiagnosed conditions.

Step	Process	Technology
1	Ingest raw doctor's notes (discharge summaries, progress notes)	MIMIC-IV dataset
2	Extract symptoms, diagnoses, medications, lab values from free text	BioBERT + medSpaCy (NER)
3	Build chronological patient health timeline	MongoDB Document Store
4	Generate patient phenotype embeddings	MongoDB Vector Search
5	Match against rare disease symptom profiles	Orphanet + HPO database
6	Display flagged diseases with confidence scores on dashboard	React.js Frontend

3. Data Sources

3.1 Patient Data — MIMIC-IV

MIMIC-IV is a **free, publicly available** dataset of real ICU patient records from **Beth Israel Deaconess Medical Center** (a Harvard-affiliated teaching hospital). Access requires completion of a CITI ethics course (~2-3 hours).

Table	Contents	Usage in Our Project
discharge.csv	Free-text discharge summaries written by doctors	PRIMARY INPUT — AI reads these notes to extract symptoms

diagnoses_icd.csv	ICD-10 diagnosis codes per patient admission	Ground truth for validating our AI's predictions
prescriptions.csv	All medications prescribed	Patient timeline + potential drug interaction flags
labevents.csv	Lab test results (blood counts, glucose, etc.)	Abnormal labs are symptoms too (e.g., low platelets)
patients.csv	Age, gender, demographics	Patient profile construction

3.2 Rare Disease Database — Orphanet + HPO

Database	What It Contains	Usage
Orphanet (en_product4.xml)	7,000+ rare diseases with associated symptom lists and prevalence	The 'answer key' — maps diseases to symptoms
HPO (phenotype.hpoa)	Standardized symptom-to-disease mappings (e.g., HP:0000958 = butterfly rash)	Bridges free-text symptoms to disease codes
HPO Ontology (hp.obo)	Hierarchy of all medical symptom terms	Standardizes AI-extracted symptoms into codes

3.3 Pre-Trained AI Models (No Training Required)

- **BioBERT** (dmis-lab/biobert-v1.1) — Pre-trained on PubMed biomedical literature. Understands medical language.
- **medSpaCy** — Clinical NLP pipeline for extracting symptoms, medications, and diagnoses from doctor's notes.
- **scispaCy** — Scientific entity recognition and medical concept linking.

4. Case Study: How RareSense.AI Detects Lupus

Below is a walkthrough of how the system would process a real-world scenario where a patient visits multiple doctors over 3 years, and no single doctor catches the underlying rare disease.

4.1 Raw Clinical Notes (Input)

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Visit 1 — Dermatologist (Jan 2023):
"28F presents with erythematous malar rash across both cheeks. Reports
photosensitivity. Prescribed topical hydrocortisone. Follow-up in 4 weeks."

Visit 2 — Hematologist (Aug 2023):
"Patient referred for thrombocytopenia. Platelet count 89,000/uL (normal:
150,000-400,000). CBC otherwise unremarkable. Monitor and repeat labs in 3 months."

Visit 3 — Nephrologist (Mar 2024):
"Urinalysis shows proteinuria (2+). eGFR mildly reduced at 72. No prior renal history.
Started ACE inhibitor. Renal biopsy considered if worsening."

Visit 4 — Primary Care (Nov 2024):
"Patient reports persistent fatigue, arthralgia in bilateral hands and knees.
Attributes to work stress. OTC ibuprofen recommended."
```

No single doctor above suspected a systemic condition. Each treated their own finding in isolation.

4.2 AI-Extracted Entities (NLP Output)

Visit	Extracted Symptoms	HPO Code
Visit 1	Malar rash (butterfly rash)	HP:0025300
Visit 1	Photosensitivity	HP:0000992
Visit 2	Thrombocytopenia (low platelets)	HP:0001873
Visit 3	Proteinuria (protein in urine)	HP:0000790
Visit 3	Reduced renal function	HP:0012622
Visit 4	Chronic fatigue	HP:0012432
Visit 4	Arthralgia (joint pain)	HP:0002829

4.3 Rare Disease Match (Output)

The system cross-references the 7 extracted phenotypes against the Orphanet/HPO database and returns:

Rank	Disease	Matched Symptoms	Confidence
1	**Systemic Lupus Erythematosus (SLE)** ORPHA:536	Malar rash + Photosensitivity + Thrombocytopenia + Proteinuria + Fatigue + Arthralgia **6 / 7 symptoms match**	**87%**
2	Antiphospholipid Syndrome ORPHA:464	Thrombocytopenia + Proteinuria + Fatigue 3 / 7 symptoms match	**41%**
3	Mixed Connective Tissue Disease ORPHA:809	Arthralgia + Fatigue + Malar rash 3 / 7 symptoms match	**38%**

Result: RareSense.AI flags **Systemic Lupus Erythematosus (SLE)** with 87% confidence. A condition that would have taken years to diagnose through traditional clinical pathways is identified in **seconds** by connecting symptoms from 4 different doctors over 2 years.

5. Technology Stack Summary

- **Database:** MongoDB (Document Store + Vector Search Index)
- **NLP Engine:** BioBERT + medSpaCy + scispaCy
- **Backend:** Python (FastAPI)
- **Frontend:** React.js (Clinician Dashboard)
- **Patient Data:** MIMIC-IV (40,000+ real ICU records)
- **Disease Data:** Orphanet (7,000+ rare diseases) + HPO

6. Next Steps (Post Review 1)

- Complete MIMIC-IV data access (CITI course + PhysioNet registration)

- Download and parse Orphanet XML + HPO annotations
- Set up NLP pipeline (medSpaCy + BioBERT) on sample discharge notes
- Design MongoDB schema for patient timelines
- Build initial matching prototype against top 200 rare diseases